Introduction to Probability concepts

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment.**
- The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as **S** .
  - A sample space is discrete if it consists of a finite or countable infinite set of outcomes.
  - A sample space is continuous if it contains an interval (either finite or infinite) of real numbers.
- An **event** is a subset of the sample space of a random experiment.

Question:

Consider the random experiment of tossing a fair coin twice.

1. Write down the **sample space** for this experiment.
2. State whether the sample space is **discrete or continuous**, and explain why.
3. Define an **event** where at least one Head appears, and list its outcomes.
4. If E_1 = "First toss is Head" and E_2 = "Second toss is Head", find $E_1 \cup E_2$ and $E_1 \cap E_2$.
5. Write the **complement** of the event "At least one Head appears".
6. Give an example of two **mutually exclusive events** in this experiment.

For the random experiment of tossing a fair coin twice,

1. Write down the **sample space** for this experiment.

$$S = \{HH, HT, TH, TT\}$$

2. State whether the sample space is **discrete or continuous**, and explain why.

The sample space is discrete, since it contains a finite number of outcomes (only 4).

- The **union** of two events is the event that consists of all outcomes that are contained in either of the two events. We denote the union as $E_1 \cup E_2$.
- The **intersection** of two events is the event that consists of all outcomes that are contained in both of the two events. We denote the intersection as $E_1 \cap E_2$.
- The **complement** of an event in a sample space is the set of outcomes in the sample space that are not in the event. We denote the complement of the event E as E' or E^c.
- Two events E_1 and E_2 are said to be **mutually exclusive events** if $E_1 \cap E_2 = \emptyset$.

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3. Define an **event** where at least one Head appears, and list its outcomes.

$$E = \{HT, TH, HH\}$$

4. If E_1= First toss is Head $= \{HT, HH\}$ and E_2 = Second toss is Head $= \{TH, HH\}$,

$$E_1 \cup E_2 = \{HT, TH, HH\}$$

$$E_1 \cap E_2 = \{HH\}$$

5. Write the **complement** of the event "At least one Head appears".

E = At least one Head appears = $\{TH, HT, HH\} \Rightarrow\ E' = \{TT\}$

6. Give an example of two **mutually exclusive events** in this experiment.

E_1 = Both tosses are Head $= \{HH\}$

E_2 = Both tosses are Tail $= \{TT\}$

$E_1 \cap E_2 = \emptyset \Rightarrow E_1, E_2$ are mutually exclusive events.

Probability of an Event

The probability of an event E is calculated by dividing the number of favorable outcomes by the total number of possible outcomes.

$$P(E) = \frac{n(E)}{n(S)}$$

Equally Likely Outcomes

Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $\frac{1}{N}$.

Axioms of Probability

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties:

If S is the sample space and E is any event in a random experiment,

- $P(S) = 1.$
- $0 \leq P(E) \leq 1.$
- For any two mutually exclusive events E_1, E_2

$$P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

Note:

- $P(\emptyset) = 0$
- For any event E, $P(E') = 1 - P(E)$.
- If the event E_1 is contained in the event E_2, $P(E_1) \leq P(E_2)$.
- For any two events E_1, E_2, $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.
- For any three events E_1, E_2, E_3,

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3).$$

- For a collection of mutually exclusive events,

$$P(E_1 \cup E_2 \cup \ldots \cup E_k) = P(E_1) + P(E_2) + \cdots + P(E_k)$$

Question:

Each of the possible five outcomes of a random experiment is equally likely. The sample space is $\{a, b, c, d, e\}$. Let A denote the event $\{a, b\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

a) $P(A)$

b) $P(B)$

c) $P(A')$

d) $P(A \cup B)$

e) $P(A \cap B)$

Each of the possible five outcomes of a random experiment is equally likely. The sample space is $\{a, b, c, d, e\}$. Let A denote the event $\{a, b\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

a) $P(A) = \frac{n(A)}{n(S)} = \frac{2}{5}$

b) $P(B) = \frac{n(B)}{n(S)} = \frac{3}{5}$

c) $P(A') = 1 - P(A) = 1 - \frac{2}{5} = \frac{3}{5}$

d) $P(A \cup B) = P(A) + P(B) = 1$

e) $P(A \cap B) = P(\emptyset) = 0$

Question:

If $P(A) = 0.3, P(B) = 0.2, P(A \cap B) = 0.1$, determine the following probabilities:

a) $P(A')$

b) $P(A \cup B)$

c) $P(A' \cap B)$

d) $P(A \cap B')$

e) $P[(A \cup B)']$

f) $P(A' \cup B)$

If $P(A) = 0.3, P(B) = 0.2, P(A \cap B) = 0.1$, determine the following probabilities:

a) $P(A') = 1 - 0.3 = 0.7$

b) $P(A \cup B) = 0.3 + 0.2 - 0.1 = 0.4$

c) $P(A' \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$

d) $P(A \cap B') = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$

e) $P[(A \cup B)'] = 1 - P(A \cup B) = 1 - 0.4 = 0.6$

f) $P(A' \cup B) = 1 - P(A \cap B') = 1 - 0.2 = 0.8$

Conditional Probability

The probability of an event B under the knowledge that the outcome will be in event A is denoted as $P(B \mid A)$ and this is called the conditional probability of B given A.

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In a deck of 52 cards where two cards are being drawn, consider the events :

- A: getting a red card on the first draw, and
- B: getting a red card on the second draw.

The conditional probability of getting a red card on the second draw B given that we got a red card on the first draw A is $P(B \mid A)$.

After drawing a red card on the first draw, there are 25 red cards and 51 cards remaining in the deck. So, $P(B \mid A) = \frac{25}{51}$.

Conditional Probability

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$$P\ (B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Similarly,

$$P\ (A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Two events are independent if any one of the following equivalent statements is true:

i. $P(A \mid B) = P(A)$

ii. $P(B \mid A) = P(B)$

iii. $P(A \cap B) = P(A)P(B)$

Baye's Theorem

If $E_1, E_2, \ldots, E_n$ are n mutually exclusive and exhaustive events and B is any event,

$$P(E_i|B) = \frac{P(E_i)P(B|E_i)}{\sum_{k=1}^{n} P(E_k)P(B|E_k)}$$

for $P(B) > 0$.

Question:

Bag 1 contains 4 white and 6 black balls while another bag 2 contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from bag 1.

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E_1 : choosing bag 1, E_2 : choosing bag 2, A : drawing a black ball

Then, $P(E_1) = P(E_2) = \frac{1}{2}$

$P(A|E_1) = P(drawing\ a\ black\ ball\ from\ bag\ 1) = \frac{6}{10} = \frac{3}{5}$

$P(A|E_2) = P(drawing\ a\ black\ ball\ from\ bag\ 2) = \frac{3}{7}$

By Baye's theorem, the probability of drawing a black ball from bag 1 out of two bags,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} = \frac{7}{12}$$

Random variable

- **A random variable** is a function that assigns a real number to each outcome in the sample space of a random experiment.
- A discrete random variable is a random variable with a finite (or countably infinite) range.
  - Example: the number of heads in 4 tosses of a coin - it cannot assume values other than 0, 1, 2, 3, 4
- A continuous random variable is a random variable with an interval (either finite or infinite) of real numbers for its range.
  - Examples : the weights (in kg) of a group of individuals, the heights of a group of individuals

Discrete Probability Distribution

Let a random variable X assumes values $x_1, x_2, x_3, \ldots, x_n$ with probabilities $p_1, p_2, p_3, \ldots, p_n$ respectively, where $P(X = x_i) = p_i \geq 0$ for each x_i and $p_1 + p_2 + \cdots + p_n = \sum_{i=1}^{n} p_i = 1$. Then

$\boldsymbol{X}$	$\boldsymbol{x_1}$	$\boldsymbol{x_2}$	$\boldsymbol{x_3}$	...	$\boldsymbol{x_n}$
$P(X)$	p_1	p_2	p_3	...	p_n

is called the discrete probability distribution for X and it says how a total probability of 1 is distributed over several values of the random variable.

Continuous Probability Distribution

If a random variable is a continuous variable, its probability distribution is called a continuous probability density function. If X a continuous random variable, then the probability density function $f(x)$ satisfies the following properties:

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

PDF can be used to determine the probabilities of the C.R.V.

i. $P(x \geq 0) = \int_0^{\infty} f(x)dx$

ii. $P(x \leq 0) = \int_{-\infty}^{0} f(x)dx$

iii. $P(a \leq x \leq b) = \int_a^b f(x)dx$

Distribution function of a discrete random variable

If X is a discrete random variable, then $P(X \leq x)$ is called the distribution function and denoted by $F(x)$.

$$F(x) = P(X \leq x) = \sum_{y;y \leq x} P(X = y)$$

For any number x, $F(x)$ is the probability that the observed value of X will be at most x.

Question:

A random variable X has the following probability function:

Values of X	0	1	2	3	4	5	6	7
$p(X)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

i. Find k,

ii. Find $P(X < 6), P(X \geq 6), P(3 < X \leq 6)$.

i. Since $\sum_{x=0}^{7} p(x) = 1$, we have

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 1 \Rightarrow (10k - 1)(k + 1) = 0$$

$$\Rightarrow \text{k} = \frac{1}{10} \qquad (\because p(x) \geq 0)$$

ii. $P(X < 6) = P(X = 0) + P(X = 1) + \cdots + P(X = 5)$

$= 0 + k + 2k + 2k + 3k + k^2$

$= 8k + k^2 = \frac{8}{10} + \frac{1}{100} = \frac{81}{100}$

$P(X \geq 6) = P(X = 6) + P(X = 7) = 2k^2 + 7k^2 + k = \frac{9}{100} + \frac{1}{10} = \frac{19}{100}$

$P(3 < X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$

$= 3k + k^2 + 2k^2 = \frac{3}{10} + \frac{3}{100} = \frac{33}{100}$

Distribution function of a continuous random variable

Let X be a continuous r.v. Then a probability distribution or probability density function (pdf) of X is a function $f(x)$ such that for any two numbers a and b with $a \neq b$,

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

That is, the probability that X takes on a value in the interval $[a, b]$ is the area above this interval and under the graph of the density function.

For $f(x)$ to be a pdf, it must satisfy the following two conditions:

i. $f(x) \geq 0$ for all x

ii. $\int_{-\infty}^{\infty} f(x)dx = area\ under\ the\ entire\ graph\ of\ f(x) = 1$

Question

The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to the center point, and let X be the angle measured clockwise to the location of an imperfection. A pdf for X is

$$f(x) = \begin{cases} \frac{1}{360}, 0 \le x < 360 \\ 0, otherwise \end{cases}$$

Find the probability that the angle is between 90^0 and 180^0.

Question

The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to the center point, and let X be the angle measured clockwise to the location of an imperfection. A pdf for X is

$$f(x) = \begin{cases} \frac{1}{360}, 0 \le x < 360 \\ 0, otherwise \end{cases}$$

Find the probability that the angle is between 90^0 and 180^0.

$$P(90 \le X \le 180) = \int_{90}^{180} \frac{1}{360} dx = \frac{x}{360}\Big]_{x=90}^{x=180} = \frac{1}{4} = 0.25$$

Cumulative Distribution Function

The cumulative distribution function of a discrete random variable X, denoted as $F(x)$, is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i).$$

The cumulative distribution function of a continuous random variable X, denoted as $F(x)$, is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} p(u)du\,, -\infty < x < \infty\,.$$

Properties:

i. $0 \leq F(x) \leq 1$.

ii. If $x \leq y$, then $F(x) \leq F(y)$. ($F(x)$ is non decreasing)

Question:

Suppose that a day's production of 850 manufactured parts contains 50 parts that do not conform to customer requirements. Two parts are selected at random, without replacement, from the batch. Let the random variable X equal the number of nonconforming parts in the sample. What is the cumulative distribution function of X?

First, let us find the probability mass function of X.

$X =$ the number of nonconforming parts in the sample = 0, 1, or 2.

$$P(X = 0) = \frac{800}{850} \cdot \frac{799}{849} = 0.886$$

$$P(X = 1) = \frac{800}{850} \cdot \frac{50}{849} + \frac{50}{850} \cdot \frac{800}{849} = 0.111$$

$$P(X = 2) = \frac{50}{850} \cdot \frac{49}{849} = 0.003$$

Thus,

x_i	0	1	2
$p(x_i)$	0.886	0.111	0.003

Therefore,

$F(0) = P(X \leq 0) = 0.886$

$F(1) = P(X \leq 1)$

$= P(X = 0) + P(X = 1)$

$= 0.886 + 0.111 = 0.997$

$F(2) = P(X \leq 2)$

$= P(X = 0) + P(X = 1) + P(X = 2)$

$= 0.886 + 0.111 + 0.003 = 1$

Expected values

Consider a university having 15,000 students and let X = the number of courses for which a randomly selected student is registered. The pmf of X follows.

x	1	2	3	4	5	6	7
$p(x)$	.01	.03	.13	.25	.39	.17	.02

Since $p(1) = 0.01$, we see that $(.01) * (15{,}000) = 5150$ of the students are registered for one course, and similarly for the other x values.

x	1	2	3	4	5	6	7
$p(x)$	.01	.03	.13	.25	.39	.17	.02
Number registered	150	450	1950	3750	5850	2550	300

The average number of courses per student, or the average value of X in the population, results from computing the total number of courses taken by all students and dividing by the total number of students.

Since each of 150 students is taking one course, these 150 contribute 150 courses to the total. Similarly, 450 students contribute 2(450) courses, and so on.

The population average value of X is then

$$\frac{1(150) + 2(450) + 3(1950) + \cdots + 7(300)}{15{,}000} = 4.57$$

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$$\frac{1(150) + 2(450) + 3(1950) + \cdots + 7(300)}{15{,}000} = 4.57$$

Another way to calculate the population average of X is

$$1 \cdot p(1) + 2 \cdot p(2) + \cdots + 7 \cdot p(7)$$

Thus, we need only the possible values of X along with their probabilities (proportions).

The average or mean value of X is then a weighted average of the possible values $1, \ldots, 7$, where the weights are the probabilities of those values.

The Expected Value (Mean) of *X*

DEFINITION

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The **expected value** or **mean value** of X, denoted by $E(X)$ or μ_X or just μ, is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

DEFINITION

The **expected** or **mean value** of a continuous rv X with pdf $f(x)$ is

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

The Variance of X

DEFINITION

Let X have pmf $p(x)$ and expected value μ. Then the **variance** of X, denoted by $V(X)$ or σ_X^2, or just σ^2, is

$$V(X) = \sum_D (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

DEFINITION

The **variance** of a continuous random variable X with pdf $f(x)$ and mean value μ is

$$\sigma_X^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x)dx = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is $\sigma_X = \sqrt{V(X)}$.

PROPOSITION

$$V(X) = E(X^2) - [E(X)]^2$$

Properties:

- $E(k) = k, k$ is a constant
- $E(aX + b) = aE(X) + b$
- $E(X^2) = \sum x^2 p(x)\ or\ \int_{-\infty}^{\infty} x^2 p(x) dx$
- $V(k) = 0, k$ is a constant
- $V(aX) = a^2 V(X)$
- $V(aX + b) = a^2 V(X)$

Question:

Let X be a number on a die when thrown. Find the mean, variance and SD of X.

Solution:

The pdf is

$\boldsymbol{X}$	1	2	3	4	5	6
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$Mean = E(X) = \sum_{x=1}^{6} xp(x) = \frac{1}{6}[1 + 2 + \cdots + 6] = \frac{7}{3}$$

$$E(X^2) = \sum_{x=1}^{6} x^2 p(x) = \frac{1}{6}[1^2 + 2^2 + \cdots + 6^2] = \frac{91}{6}$$

$$Variance = V(X) = E(X^2) - [E(X)]^2 = \frac{91}{6} - \left(\frac{7}{3}\right)^2 = \frac{175}{18}$$

$$SD = \sqrt{V(X)} = \sqrt{\frac{175}{18}}$$

Jointly Distributed Random Variables

- There are many experimental situations in which more than one random variable (rv) will be of interest to an investigator.
- The probability mass function (pmf) of a single discrete rv X specifies how much probability mass is placed on each possible X value.
- The joint pmf of two discrete rv's X and Y describes how much probability mass is placed on each possible pair of values (x, y).

Joint distribution of two discrete random variables

DEFINITION

Let X and Y be two discrete rv's defined on the sample space $\mathcal{S}$ of an experiment. The **joint probability mass function** $p(x, y)$ is defined for each pair of numbers (x, y) by

$$p(x, y) = P(X = x \text{ and } Y = y)$$

It must be the case that $p(x, y) \geq 0$ and $\sum_x \sum_y p(x, y) = 1$.

Now let A be any particular set consisting of pairs of (x, y) values (e.g., $A = \{(x, y): x + y = 5\}$ or $\{(x, y): \max(x, y) \leq 3\}$). Then the probability $P[(X, Y) \in A]$ that the random pair (X, Y) lies in the set A is obtained by summing the joint pmf over pairs in A:

$$P[(X, Y) \in A] = \sum_{(x,y) \in A} \sum p(x, y)$$

Joint distribution of two continuous random variables

DEFINITION

Let X and Y be continuous rv's. A **joint probability density function** $f(x, y)$ for these two variables is a function satisfying $f(x, y) \geq 0$ and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$. Then for any two-dimensional set A

$$P[(X, Y) \in A] = \iint_A f(x, y)\, dx\, dy$$

In particular, if A is the two-dimensional rectangle $\{(x, y): a \leq x \leq b, c \leq y \leq d\}$, then

$$P[(X,Y) \in A] = P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

Question:

A bank operates both a drive-up facility and a walk-up window. On a randomly selected day, let X be the proportion of time that the drive-up facility is in use (at least one customer is being served or waiting to be served) and Y be the proportion of time. Then the set of possible values for (X, Y) is the rectangle $D = \{(x, y): 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Suppose

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, otherwise \end{cases}.$$

a) Verify whether $f(x, y)$ can serve as a valid joint probability density function (pdf) for (X, Y).

b) If it is valid, determine the probability that neither facility is busy more than one-quarter of the time.

To verify that this is a legitimate pdf, note that $f(x, y) \geq 0$ and

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\,dx\,dy &= \int_0^1\int_0^1 \frac{6}{5}(x + y^2)\,dx\,dy \\ &= \int_0^1\int_0^1 \frac{6}{5}x\,dx\,dy + \int_0^1\int_0^1 \frac{6}{5}y^2\,dx\,dy \\ &= \int_0^1 \frac{6}{5}x\,dx + \int_0^1 \frac{6}{5}y^2\,dy = \frac{6}{10} + \frac{6}{15} = 1\end{aligned}$$

$$\begin{aligned}P\left(0 \leq X \leq \frac{1}{4}, 0 \leq Y \leq \frac{1}{4}\right) &= \int_0^{1/4}\int_0^{1/4} \frac{6}{5}(x + y^2)\,dx\,dy \\ &= \frac{6}{5}\int_0^{1/4}\int_0^{1/4} x\,dx\,dy + \frac{6}{5}\int_0^{1/4}\int_0^{1/4} y^2\,dx\,dy \\ &= \frac{6}{20}\cdot\frac{x^2}{2}\Big|_{x=0}^{x=1/4} + \frac{6}{20}\cdot\frac{y^3}{3}\Big|_{y=0}^{y=1/4} = \frac{7}{640} \\ &= .0109\end{aligned}$$

Question

Show that the following function satisfies the properties of a joint probability mass function.

x	y	$f_{XY}(x, y)$
1	1	1/4
1.5	2	1/8
1.5	3	1/4
2.5	4	1/4
3	5	1/8

Determine the following:

(a) $P(X < 2.5, Y < 3)$

(b) $P(X < 2.5)$

(c) $P(Y < 3)$

(d) $P(X > 1.8, Y > 4.7)$

Solution:

(a) $P(X < 2.5, Y < 3)$	$\frac{3}{8}$
(b) $P(X < 2.5)$	$\frac{5}{8}$
(c) $P(Y < 3)$	$\frac{3}{8}$
(d) $P(X > 1.8, Y > 4.7)$	$\frac{1}{8}$

Marginal probability distribution

- If more than one random variable is defined in a random experiment, it is important to distinguish between the joint probability distribution of X and Y and the probability distribution of each variable individually.
- The individual probability distribution of a random variable is referred to as its **marginal probability distribution.**
- In general, the marginal probability distribution of X can be determined from the joint probability distribution of X and other random variables.

Consider discrete random variables X and Y.

To determine $P(X = x)$, we sum $P(X = x, Y = y)$ over all points in the range of (X, Y) for which $X = x$.

$$f_X(x) = \sum_y P(X = x, Y = y)$$

Calls are made to check the airline schedule at your departure city. You monitor the number of bars of signal strength on your cell phone and the number of times you have to state the name of your departure city before the voice system recognizes the name.

In the first four bits transmitted, let

- X denote the number of bars of signal strength on your cell phone
- Y denote the number of times you need to state your departure city

The joint probability distribution function of X and Y is

	x = number of bars of signal strength		
y = number of times city name is stated	1	2	3
4	0.15	0.1	0.05
3	0.02	0.1	0.05
2	0.02	0.03	0.2
1	0.01	0.02	0.25

Find the marginal distributions of X and Y.

	x = number of bars of signal strength			
y = number of times city name is stated	1	2	3	Marginal probability distribution of Y
4	0.15	0.1	0.05	0.3
3	0.02	0.1	0.05	0.17
2	0.02	0.03	0.2	0.25
1	0.01	0.02	0.25	0.28
	0.2	0.25	0.55	
	Marginal probability distribution of X			

Marginal Probability Density Function

If the joint probability density function of random variables X and Y is $f_{XY}(x, y)$, the **marginal probability density functions** of X and Y are

$$f_X(x) = \int_y f_{XY}(x, y)\, dy \quad \text{and} \quad f_Y(y) = \int_x f_{XY}(x, y)\, dx \tag{5-3}$$

where the first integral is over all points in the range of (X, Y) for which $X = x$ and the second integral is over all points in the range of (X, Y) for which $Y = y$.

A probability for only one random variable, say, for example, $P(a < X < b)$, can be found from the marginal probability distribution of X or from the integral of the joint probability distribution of X and Y as

$$P(a < X < b) = \int_a^b f_X(x)dx = \int_a^b \left[\int_{-\infty}^{\infty} f(x, y)dy \right] dx = \int_a^b \int_{-\infty}^{\infty} f(x, y)dydx$$

Let the random variable X denote the time until a computer server connects to your machine (in milliseconds), and let Y denote the time until the server authorizes you as a valid user (in milliseconds). Each of these random variables measures the wait from a common starting time and $X < Y$. Assume that the joint probability density function for X and Y is

$$f_{XY}(x, y) = 6 \times 10^{-6} \exp(-0.001x - 0.002y) \qquad \text{for } x < y$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)\,dy\,dx = \int_{0}^{\infty}\left(\int_{x}^{\infty} 6\times 10^{-6}e^{-0.001x-0.002y}\,dy\right)dx$$

$$= 6\times 10^{-6}\int_{0}^{\infty}\left(\int_{x}^{\infty} e^{-0.002y}\,dy\right)e^{-0.001x}\,dx$$

$$= 6\times 10^{-6}\int_{0}^{\infty}\left(\frac{e^{-0.002x}}{0.002}\right)e^{-0.001x}\,dx$$

$$= 0.003\left(\int_{0}^{\infty} e^{-0.003x}\,dx\right)$$

$$= 0.003\left(\frac{1}{0.003}\right) = 1$$

$$P(X \leq 1000, Y \leq 2000) = \int_0^{1000} \int_x^{2000} f_{XY}(x, y)\, dy\, dx$$

$$= 6 \times 10^{-6} \int_0^{1000} \left(\int_x^{2000} e^{-0.002y}\, dy \right) e^{-0.001x}\, dx$$

$$= 6 \times 10^{-6} \int_0^{1000} \left(\frac{e^{-0.002x} - e^{-4}}{0.002} \right) e^{-0.001x}\, dx$$

$$= 0.003 \int_0^{1000} e^{-0.003x} - e^{-4} e^{-0.001x}\, dx$$

$$= 0.003 \left[\left(\frac{1 - e^{-3}}{0.003} \right) - e^{-4} \left(\frac{1 - e^{-1}}{0.001} \right) \right]$$

$$= 0.003(316.738 - 11.578) = 0.915$$

$$f_Y(y) = \int_0^y 6 \times 10^{-6} e^{-0.001x - 0.002y}\, dx$$

$$= 6 \times 10^{-6} e^{-0.002y} \int_0^y e^{-0.001x}\, dx$$

$$= 6 \times 10^{-6} e^{-0.002y} \left(\frac{e^{-0.001x}}{-0.001} \Big|_0^y \right)$$

$$= 6 \times 10^{-6} e^{-0.002y} \left(\frac{1 - e^{-0.001y}}{0.001} \right)$$

$$= 6 \times 10^{-3} e^{-0.002y} (1 - e^{-0.001y}) \quad \text{for } y > 0$$

$$P(Y > 2000) = 6 \times 10^{-3} \int_{2000}^{\infty} e^{-0.002y}(1 - e^{-0.001y})\, dy$$

$$= 6 \times 10^{-3} \left[\left(\frac{e^{-0.002y}}{-0.002} \Big|_{2000}^{\infty} \right) - \left(\frac{e^{-0.003y}}{-0.003} \Big|_{2000}^{\infty} \right) \right]$$

$$= 6 \times 10^{-3} \left[\frac{e^{-4}}{0.002} - \frac{e^{-6}}{0.003} \right] = 0.05$$

Theory of Estimation

- The process of drawing conclusions about a population based on information collected from the sample taken from the population is called statistical inference.
- It is divided into two: Estimation of parameters and testing of hypothesis.
- Estimation is the method of obtaining the value of an unknown parameter using the observations taken from the sample of the population.
  - Point estimation
  - Interval estimation

Point Estimation:

- When we estimate the value of an unknown parameter by a single value, it is called point estimation.
  - Example: Sample mean $(\bar{x})$ is a point estimation of population mean (μ).

Interval Estimation:

- Interval estimation is a method under which two values are estimated regarding the parameter which provides the limits within which actual value lies. These two limits gives an interval and this interval is called confidence interval or confidence limits.

- An estimate whose value lies nearer to the true value of the parameter to be estimated is called best estimate.
- A sample statistic that is used to estimate the unknown parameter is called an estimator.

Characteristics of a good estimator

✓ **Un-biased**

- An estimator $\bar{\theta}$ is said to be an unbiased estimator if the expectation of the estimator is equal to the corresponding population parameter θ.

✓**Consistent (Stability)**

- An estimator $\bar{\theta}$ is said to be consistent if the difference $\bar{\theta} - \theta$ becomes smaller as the sample size grows larger. $(\bar{\theta} \rightarrow \theta)$

✓**Efficient**

- An estimator $\bar{\theta}$ is efficient if it has the minimum variance among unbiased estimators.

✓**Sufficient**

- An estimator $\bar{\theta}$ is sufficient if it gives as much as the possible information regarding the parameter.

Recall: Baye's Theorem

If $E_1, E_2, \ldots, E_n$ are n mutually exclusive and exhaustive events and B is any event,

$$P(E_i|B) = \frac{P(E_i)P(B|E_i)}{\sum_{k=1}^{n} P(E_k)P(B|E_k)}$$

for $P(B) > 0$.

Bayesian Estimation of Parameters

- The Bayesian approach of statistical inference that combines sample information with other information that may be available prior to collecting the sample.
- Suppose that the random variable X has a probability distribution that is a function of one parameter θ.
- We write this probability distribution as $f(x|\theta)$. This notation implies that the exact form of the distribution of X is conditional on the value assigned to θ.
- The classical approach to estimation would consist of taking a random sample of size n from this distribution and then substituting the sample values x_i into the estimator for θ.

- Suppose that we have some additional information about θ and that we can summarize that information in the form of a probability distribution for , say,$f(\theta)$.
- This probability distribution is often called the **prior distribution** for θ, and suppose that the mean of the prior is μ_0 and the variance is σ_0^2 (we are now viewing the parameter as a random variable).
- The Bayesian approach to estimation uses the prior distribution for θ, $f(\theta)$, and the joint probability distribution of the sample, $f(x_1, x_2, \ldots, x_n|\theta)$ to find a **posterior distribution** for θ, say, $f(\theta|x_1, x_2, \ldots, x_n)$.
- This posterior distribution contains information from both the sample and the prior distribution for θ.

The joint probability distribution of the sample $X_1, X_2, \ldots, X_n$ and the parameter θ (remember that θ is a random variable) is

$$f(x_1, x_2, \ldots, x_n, \theta) = f(x_1, x_2, \ldots, x_n \mid \theta) f(\theta)$$

and the marginal distribution of $X_1, X_2, \ldots, X_n$ is

$$f(x_1, x_2, \ldots, x_n) = \begin{cases} \sum_{\theta} f(x_1, x_2, \ldots, x_n, \theta), & \theta \text{ discrete} \\ \int_{-\infty}^{\infty} f(x_1, x_2, \ldots, x_n, \theta)\, d\theta, & \theta \text{ continuous} \end{cases}$$

Therefore, the desired distribution is

$$f(\theta \mid x_1, x_2, \ldots, x_n) = \frac{f(x_1, x_2, \ldots, x_n, \theta)}{f(x_1, x_2, \ldots, x_n)}$$

- We define the Bayes estimator of θ as the value of $\hat{\theta}$ that corresponds to the mean of the posterior distribution $f(\theta|x_1, x_2, \ldots, x_n)$.

Example: Estimating the Probability of Heads (Coin toss)

- We want to estimate

$$\theta = probability\ of\ getting\ heads$$

based on on experimental data — say, tossing a coin 10 times and getting 7 heads.

- Instead of just saying, **"it is $\frac{7}{10}$"**, we'll use Bayesian reasoning to find how likely each possible value of θ is after seeing the data.

Step 1: Define the Random Variables

- $X =$ number of heads in 10 tosses
- $\theta =$ true (but unknown) probability of heads

Our data are summarized as:

$$x = 7 \text{ heads out of } 10$$

We want $f(\theta|x)$:

the **posterior distribution** — what we now believe about θ after seeing the data.

Step 2: Bayes' Theorem

$$f(\theta|x) = \frac{f(x|\theta)\, f(\theta)}{f(x)}$$

- $f(\theta)$: **prior** (belief before seeing data)
- $f(x|\theta)$: **likelihood** (how likely data are for a given θ)
- $f(x)$: **marginal** or normalizing constant

Step 3: Choose the Prior $f(\theta)$

Suppose before tossing the coin, we have **no reason** to believe the coin is biased.

So, we use a **Uniform(0,1)** prior:

$$f(\theta) = 1, \quad 0 \leq \theta \leq 1$$

That means *all values* between 0 and 1 are equally likely.

Step 4: Write the Likelihood function $f(x|\theta)$

The number of heads in 10 tosses follows a **Binomial distribution**:

$$f(x|\theta) = \binom{10}{x}\theta^x(1-\theta)^{10-x}$$

For $x = 7$:

$$f(7|\theta) = \binom{10}{7}\theta^7(1-\theta)^3$$

Step 5: Compute the Joint Distribution $f(x, \theta)$

Using:

$$f(x, \theta) = f(x|\theta) f(\theta)$$

Since $f(\theta) = 1$:

$$f(x, \theta) = \binom{10}{7} \theta^7 (1 - \theta)^3$$

This gives the combined probability (or intersection) of seeing both $x = 7$ and a given θ.

Step 6: Compute the Marginal Distribution $f(x)$

We integrate over all possible θ:

$$f(x) = \int_0^1 f(x, \theta)\, d\theta$$

$$= \binom{10}{7} \int_0^1 \theta^7 (1-\theta)^3\, d\theta$$

The integral $\int_0^1 \theta^7 (1-\theta)^3 d\theta = B(8, 4)$,
where B is the **Beta function**.

Thus:

$$f(x) = \binom{10}{7} B(8, 4)$$

Step 7: Write the Posterior Distribution

Now plug values into Bayes' theorem:

$$f(\theta|x) = \frac{f(x,\theta)}{f(x)} = \frac{\binom{10}{7}\theta^7(1-\theta)^3}{\binom{10}{7}B(8,4)}$$

Simplify:

$$f(\theta|x) = \frac{1}{B(8,4)}\theta^7(1-\theta)^3$$

$\boldsymbol{f(\theta|x)}$is called the $Beta(8,4)$ distribution.

So our posterior belief is:

$$\boxed{f(\theta|x) = \text{Beta}(8, 4)}$$

The **mean** (expected value) of a Beta(α, β) distribution is:

$$E[\theta|x] = \frac{\alpha}{\alpha + \beta} = \frac{8}{8 + 4} = 0.667$$

Hence, our **Bayesian estimate** for the probability of heads is:

$$\boxed{\hat{\theta} = 0.667}$$

Suppose that X is a normal random variable with unknown mean μ and known variance σ^2. The prior distribution for μ is a uniform distribution defined over the interval $[a, b]$.

(a) Find the posterior distribution for μ.

(b) Find the Bayes estimator for μ.

Since $X \sim N(\mu, \sigma^2)$

$$f(x|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This tells us how likely a data value x is, given a particular μ.

Given that μ is uniformly distributed over $[a,b]$:

$$f(\mu) = \begin{cases} \frac{1}{b-a}, a \leq \mu \leq b, \\ 0, otherwise \end{cases}$$

This shows all μ in $[a,b]$ are equally likely.

The joint distribution is

$$f(x,\mu) = f(x|\mu) * f(\mu)$$

Hence,

$$f(x,\mu) = \frac{1}{\sqrt{2\pi\sigma^2}(b-a)} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, a \leq \mu \leq b.$$

The marginal distribution $f(x)$ is given by

$$f(x) = \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}(b-a)} e^{-\frac{(x-\mu)^2}{2\sigma^2}} d\mu \ .$$

$f(x)$ tells us how likely it is to observe the data value x, given the range of possible means $[a,b]$.

Therefore, the **posterior** distribution is

$$f(\mu|x) = \frac{f(x|\mu)f(\mu)}{f(x)}.$$

Let X = the time it takes a read/write head to locate a desired record on a computer disk memory device once the head has been positioned over the correct track. If the disks rotate once every 25 millisec, a reasonable assumption is that X is uniformly distributed on the interval [0, 25].

a. Compute $P(10 \leq X \leq 20)$.

b. Compute $P(X \geq 10)$.

A restaurant serves three fixed-price dinners costing \$12, \$15, and \$20. For a randomly selected couple dining at this restaurant, let X = the cost of the man's dinner and Y = the cost of the woman's dinner. The joint pmf of X and Y is given in the following table:

$p(x, y)$		y = 12	15	20
	12	.05	.05	.10
x	15	.05	.10	.35
	20	0	.20	.10

a. Compute the marginal pmf's of X and Y.

b. What is the probability that the man's and the woman's dinner cost at most \$15 each?

- Show that the following function satisfies the properties of a joint probability mass function.

x	y	$f_{XY}(x, y)$
−1	−2	1/8
−0.5	−1	1/4
0.5	1	1/2
1	2	1/8

Determine the following:
(a) $P(X < 0.5, Y < 1.5)$
(b) $P(X < 0.5)$
(c) $P(Y < 1.5)$
(d) $P(X > 0.25, Y < 4.5)$

A die is tossed thrice. A success is “getting 1 or 6 on a toss”. Find the mean and the variance of the number of successes.